

6 Plants respond to external stimuli.

(a) Phytochrome molecules are found in plants.

Phytochrome exists in two forms, Pr (P_{660}) and Pfr (P_{730}).

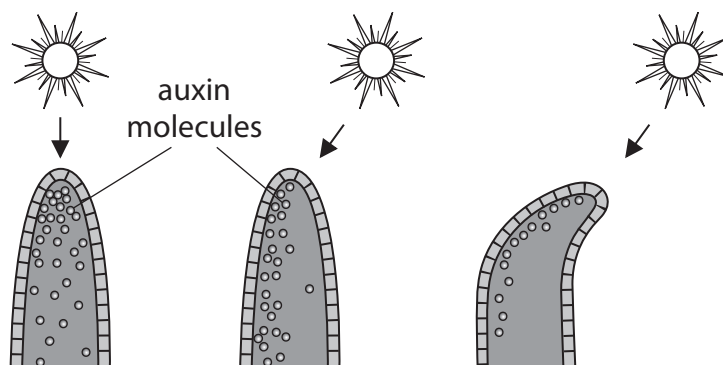
Which row shows the action of these two forms of phytochrome?

(1)

		Absorbs far red light	Breaks down in the dark
☐	A	Pfr	Pfr
☐	B	Pfr	Pr
☐	C	Pr	Pfr
☐	D	Pr	Pr

(b) Auxins (IAA) are plant growth substances.

The diagram shows auxin molecules producing a phototropic response in a shoot.



Explain the role of auxins in this phototropic response.

Use the information in the diagram to support your answer.

(3)

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- (c) Seeds contain starch. This starch is used to supply energy when the seed germinates.

Gibberellins are a group of naturally occurring plant growth substances.

Gibberellins regulate starch hydrolysis in the seed.

Explain how gibberellins can produce this effect on the seed.

(4)

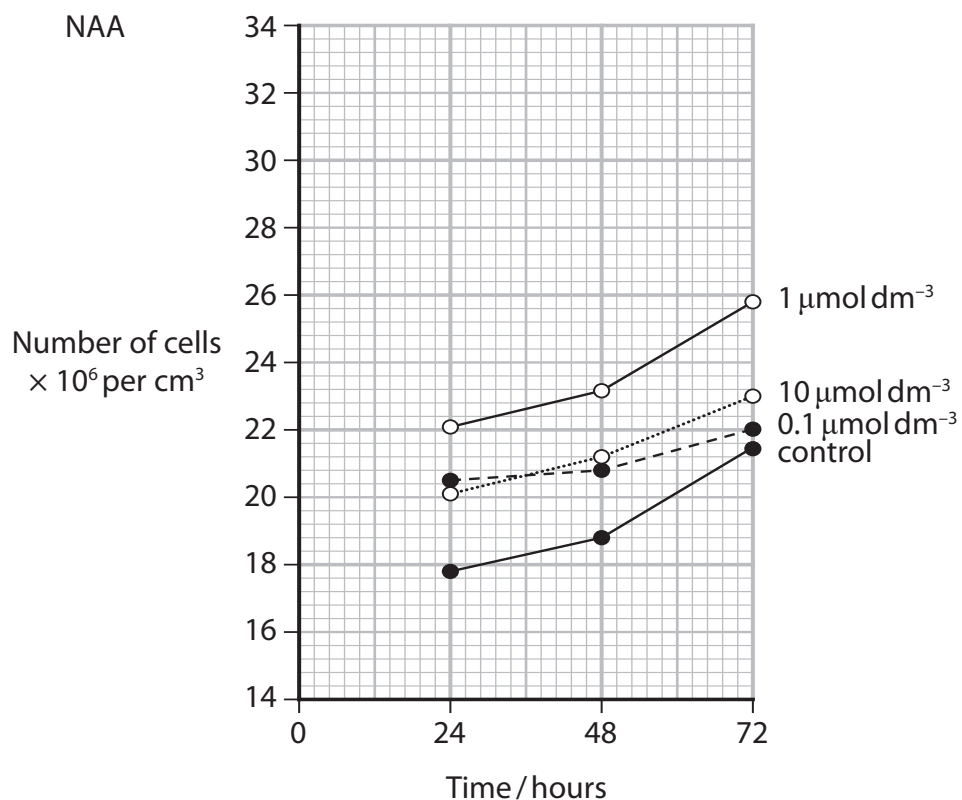
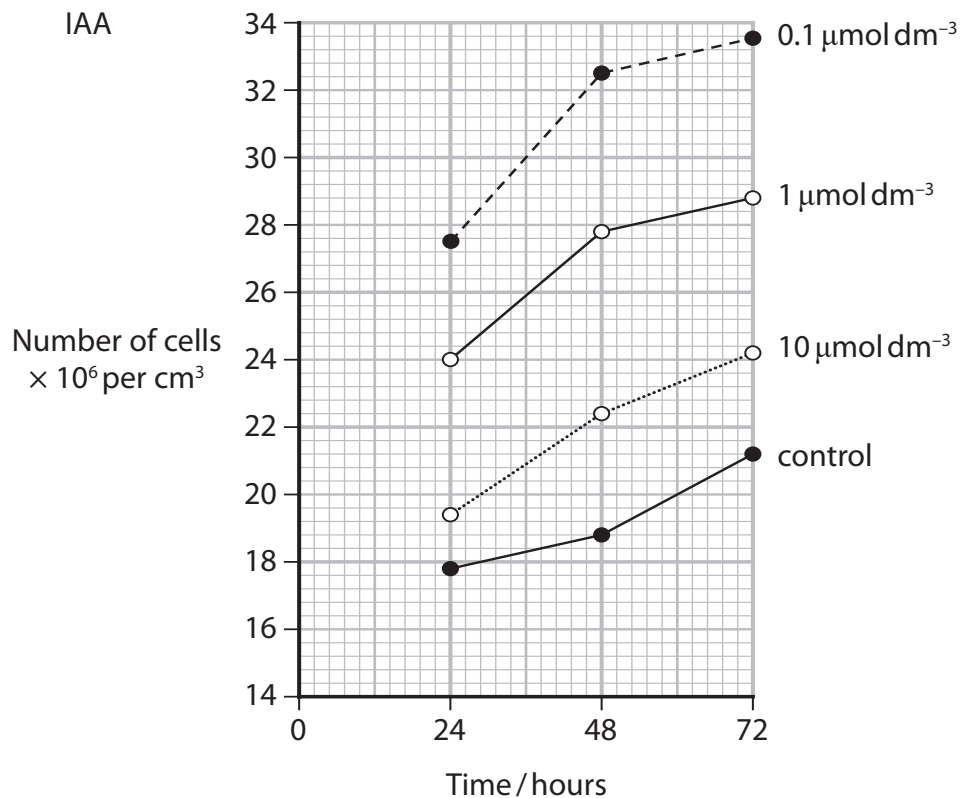
[illegible]

(d) Indole acetic acid (IAA) is a natural plant auxin. 1-Naphthaleneacetic acid (NAA) is a synthetic auxin.

The effects of these two auxins on the growth of *Chlorella* was investigated.

Chlorella is a single-celled photosynthetic organism.

The graphs show the effect of different concentrations of the auxins on the growth of *Chlorella*.



Comment on the effect of these auxins on the growth of *Chlorella*.

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(Total for Question 6 = 12 marks)